In[ ]:=
```
S[x_, n_] := Sqrt[(n - 1) / n + x x / n]
DHC5[x_, n_, Kp_, sn_] := sn (1 / Sqrt[n (n + 1)] x + (S[x, n] - 1) Kp)
DAF[x_, n_, Kp_, sn_, kn_, knn_] := sn (S[x, n] (Kp + knn) - (Kp + kn))
DPNEC[x_, n_, Kp_, sn_, kn_, knn_] :=
 DHC5[x, n, Kp, sn] - DAF[x, n, Kp, sn, kn, knn]
```

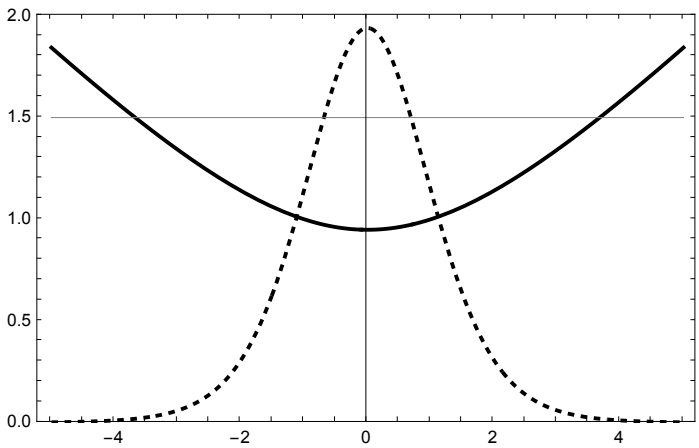
Sn Ratio

In[ ]:=
```
SS = 10 (* sample size *)
ai = Plot[S[x, SS], {x, -5, 5},
   PlotStyle → GrayLevel[0], PlotRange → {0, 2}, Frame → True];
bi = Plot[5 PDF[StudentTDistribution[SS - 1], x], {x, -5, 5},
   PlotStyle → {Dashing[{0.01, 0.01}], GrayLevel[0]}, Frame → True];
ci = Plot[1.5, {x, -5, 5}, PlotStyle → {GrayLevel[0.5], Thickness[0.001]}];
Show[ai, bi, ci]
```

Out[ ]=

10

Out[ ]=



intersections between curves and z (=log10(v))

In[ ]:=
```
solX = Solve[S[x, n] == z, x]
minSn = S[0, n]
```

Out[ ]=

$$\left\{\left\{x \to -\sqrt{1 - n + n z^2}\right\}, \left\{x \to \sqrt{1 - n + n z^2}\right\}\right\}$$

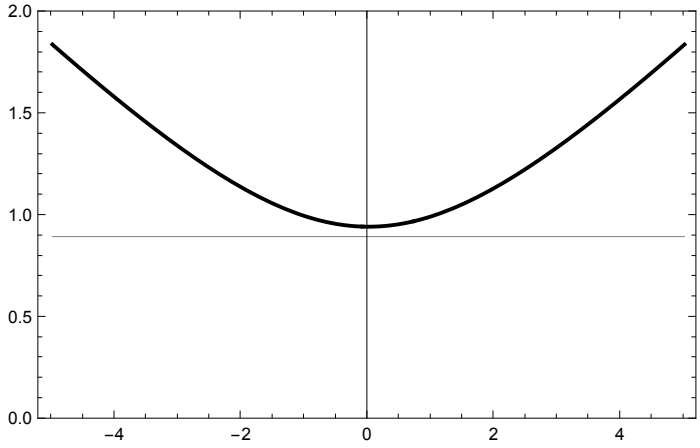
Out[ ]=

$$\sqrt{\frac{-1 + n}{n}}$$

if z < minSn, then there are no intersections. For example, when SS = 10, minSn is 0.948683 (snn will never be lower than 0.95 sn). Horizontal gray line has no intersections with thick curve for S(x)

In[ ]:=
```
ai = Plot[S[x, SS], {x, -5, 5},
   PlotStyle → GrayLevel[0], PlotRange → {0, 2}, Frame → True];
bi = Plot[0.9, {x, -5, 5}, PlotStyle → {GrayLevel[0.5], Thickness[0.001]}];
Show[ai, bi]
```

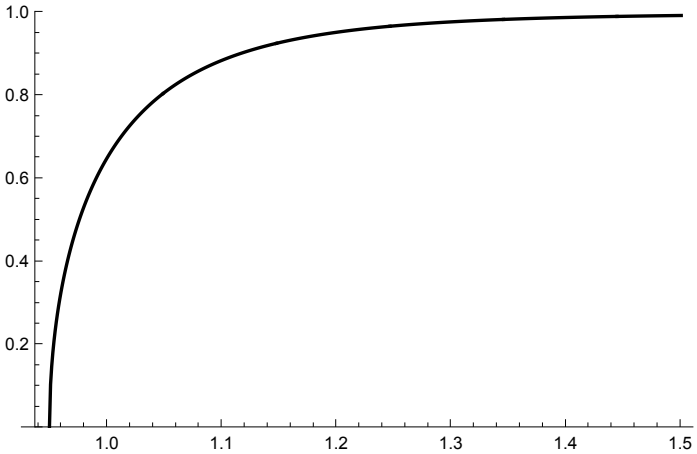
Out[ ]=



CDF

In[ ]:=
```
SS = 10;
minSn = S[0, n];
del = 0.001;
intRng = x /. solX /. n → SS;
znow = minSn /. n → SS;
myCDF = {{znow, 0}};

Do[
  znow = znow + del;
  If[znow > 1.5, Break[]];
  xl = intRng[[1]] /. z → znow;
  xh = -xl;
  p =
   CDF[StudentTDistribution[SS - 1], xh] - CDF[StudentTDistribution[SS - 1], xl];
  AppendTo[myCDF, {znow, p}];
  , {k, 10 000}];
grCDF = ListPlot[myCDF, Joined → True,
  PlotRange → {0, 1}, PlotStyle → {Thickness[0.005], GrayLevel[0]}]
```

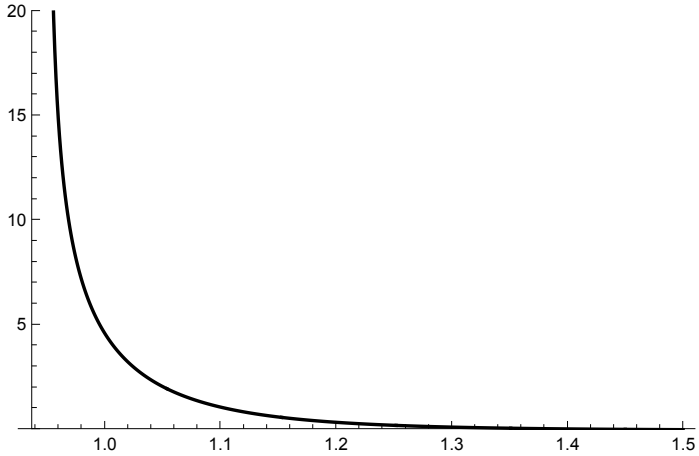
Out[ ]=



PDF

In[ ]:=
```
ya = Transpose[myCDF][[2]];
xa = Transpose[myCDF][[1]];
myPDF = {};
Do[
  AppendTo[myPDF, {xa[[k]], (ya[[k + 1]] - ya[[k]]) / (xa[[k + 1]] - xa[[k]])}];
  , {k, Length[myCDF] - 1}];
grPDF = ListPlot[myPDF, Joined → True,
  PlotRange → {0, 20}, PlotStyle → {Thickness[0.005], GrayLevel[0]}]
```

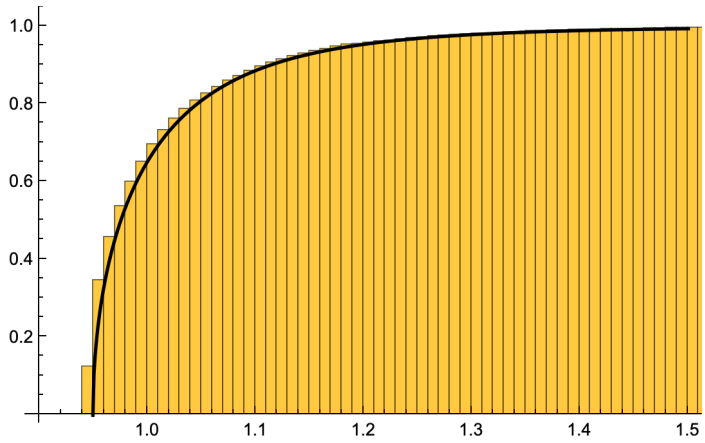
Out[ ]=



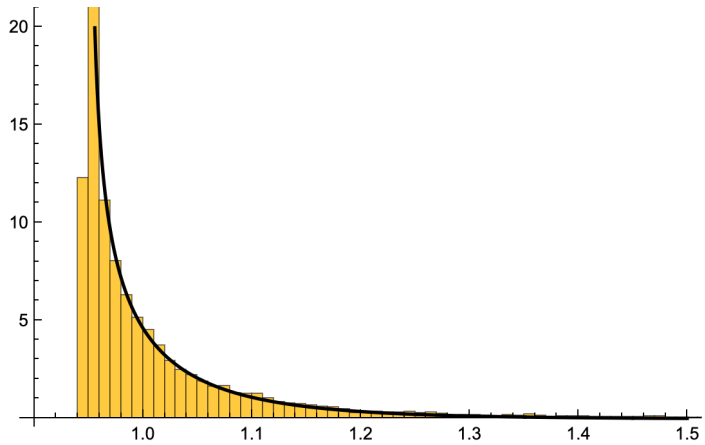
comparison with monte-carlo

In[ ]:=
```
snRatios = {};
Do[
  preSample = Table[RandomReal[NormalDistribution[0, 3.14]], {k, 10}];
  preSd = StandardDeviation[preSample];

  postSample = AppendTo[preSample, RandomReal[NormalDistribution[0, 3.14]]];
  postSd = StandardDeviation[postSample];

  sRatio = postSd / preSd;
  AppendTo[snRatios, sRatio];
  , {t, 10 000}];
ci = Histogram[snRatios, 100, "CDF"];
di = Histogram[snRatios, 100, "PDF"];
Show[ci, grCDF, PlotRange → {{0.9, 1.5}, {0, 1}}, AxesOrigin → {0.9, 0}]
Show[di, grPDF, PlotRange → {{0.9, 1.5}, {0, 20}}, AxesOrigin → {0.9, 0}]
```

Out[ ]=



Out[ ]=



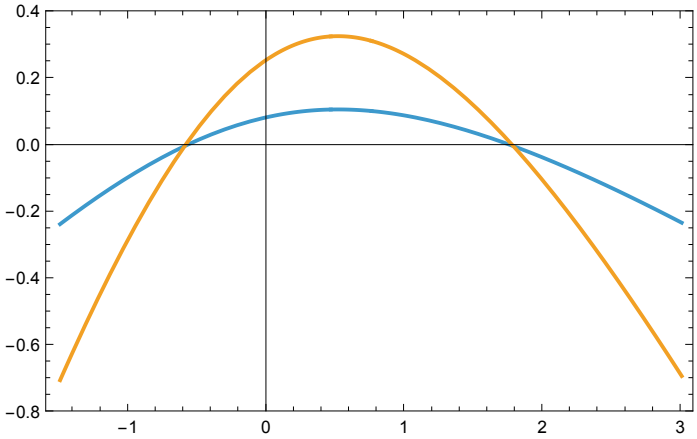
change in HC5

In[ ]:=
```
K5 = Quantile[NormalDistribution[0, 1], 0.05]
```
Out[ ]=
```
-1.64485
```

functional form of dlogHC5. sn = 0.5 and 1.5, as example

In[ ]:=
```
SS = 5;
Plot[Evaluate[{DHC5[x, SS, K5, 0.5], DHC5[x, SS, K5, 1.5]}],
 {x, -1.5, 3}, Frame → True, PlotRange → {-0.8, 0.4}]
```

Out[ ]=



obtaining the probability that post-HC5 > v pre-HC5 (z = log10(v)). n=5, v = 2, sn=1.5 as example

In[ ]:=
```
solX = FullSimplify[x /. Solve[DHC5[x, n, Kp, sn] == z, x]]
maxX = FullSimplify[x /. Solve[D[DHC5[x, n, Kp, sn], x] == 0, x][[2]]]
maxHC5 = FullSimplify[DHC5[maxX, n, Kp, sn]]

SS = 5;
params = {Kp → K5, n → SS, sn → 1.5};
If[(maxHC5 /. params) < Log10[2], Print["No solution"]]
intRng = solX /. params /. z → Log10[2];
Print["Probability that post-HC5 > 2 pre-HC5"]
(CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
  CDF[StudentTDistribution[SS - 1], intRng[[1]]])
```

Out[ ]=

$$\left\{-\frac{1}{\left(-1+Kp^2\,(1+n)\right)\,sn^2}\left(\sqrt{n\,(1+n)}\;sn\,(Kp\,sn+z)+\sqrt{Kp^2\,(1+n)\,sn^2\left(\left(-1+n+Kp^2\,(1+n)\right)sn^2+2\,Kp\,n\,(1+n)\,sn\,z+n\,(1+n)\,z^2\right)}\right),\ \frac{1}{\left(-1+Kp^2\,(1+n)\right)\,sn^2}\left(-\sqrt{n\,(1+n)}\;sn\,(Kp\,sn+z)+\sqrt{Kp^2\,(1+n)\,sn^2\left(\left(-1+n+Kp^2\,(1+n)\right)sn^2+2\,Kp\,n\,(1+n)\,sn\,z+n\,(1+n)\,z^2\right)}\right)\right\}$$

Out[ ]=

$$\frac{\sqrt{-1+n}}{\sqrt{-1+Kp^2\,(1+n)}}$$

Out[ ]=

$$\left(\frac{\sqrt{-1+n}}{\sqrt{n\,(1+n)}\;\sqrt{-1+Kp^2\,(1+n)}}+Kp\left(-1+\sqrt{\frac{Kp^2\left(-1+n^2\right)}{n\left(-1+Kp^2\,(1+n)\right)}}\right)\right)sn$$

Probability that post-HC5 > 2 pre-HC5

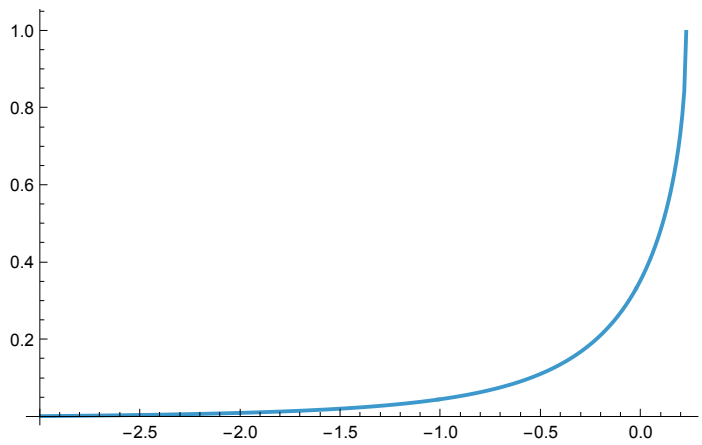
Out[ ]=

0.212828

CDF of delta log HC5, sn=1, n=5 as example (probabilities that post-HC5 is lower than pre-HC5)

In[ ]:=
```
SS = 5;
params = {Kp → K5, n → SS, sn → 1};
maxZ = maxHC5 /. params;
del = 0.01;
znow = -3;
myCDF = {};
Do[
 intRng = solX /. params /. z → znow;
 If[znow > maxZ, Break[]];
 p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
 AppendTo[myCDF, {znow, 1 - p}];
 znow = znow + del;
 , {k, 100 000}]
myCDF = AppendTo[myCDF, {maxZ, 1}];
grCDF = ListPlot[myCDF, Joined → True, PlotRange → All, AxesOrigin → {-3, 0}]
```

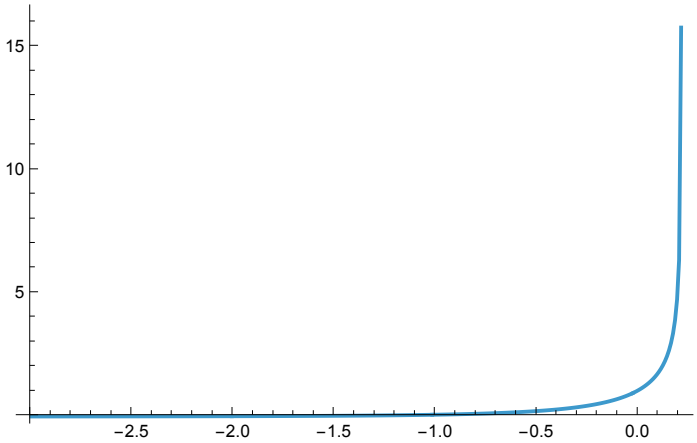
Out[ ]=



PDF

In[ ]:=
```
ya = Transpose[myCDF][[2]];
xa = Transpose[myCDF][[1]];
myPDF = {};
Do[
  AppendTo[myPDF, {xa[[k]], (ya[[k + 1]] - ya[[k]]) / (xa[[k + 1]] - xa[[k]])}];
  , {k, Length[myCDF] - 1}];
grPDF = ListPlot[myPDF, Joined → True, PlotRange → All, AxesOrigin → {-3, 0}]
```

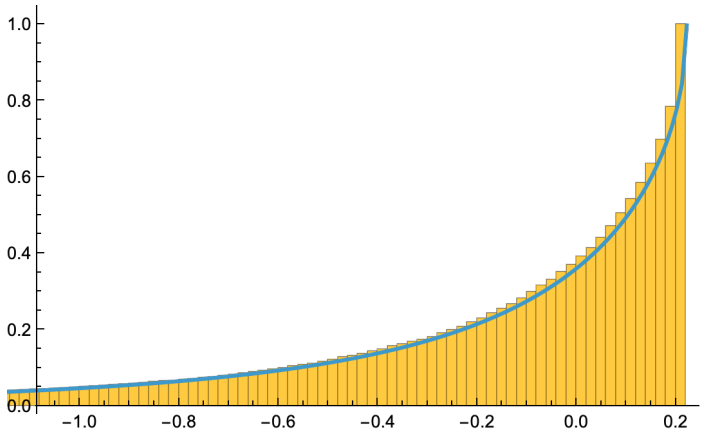
Out[ ]=



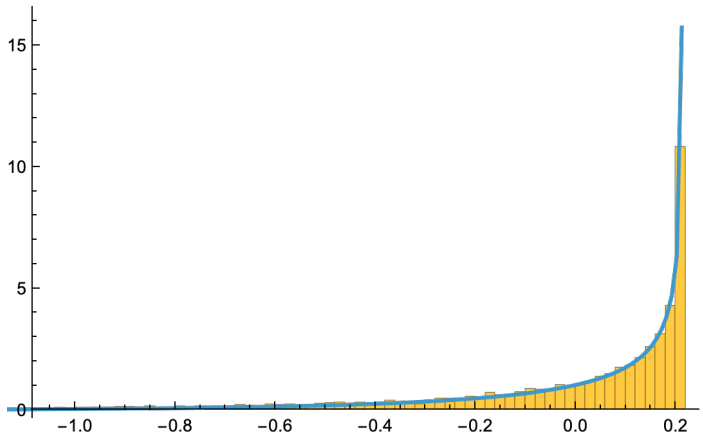
comparison with monte-carlo

In[ ]:=
```
dHC5s = {};
Do[
  preSample = Table[RandomReal[NormalDistribution[0, 3.14]], {k, SS}];
  preSd = StandardDeviation[preSample];
  preMean = Mean[preSample];
  preHC5 = preMean + K5 preSd;

  postSample = AppendTo[preSample, RandomReal[NormalDistribution[0, 3.14]]];
  postSd = StandardDeviation[postSample];
  postMean = Mean[postSample];
  postHC5 = postMean + K5 postSd;

  dHC5 = (postHC5 - preHC5) / preSd;
  AppendTo[dHC5s, dHC5];
  , {t, 10 000}];
ci = Histogram[dHC5s, 500, "CDF"];
di = Histogram[dHC5s, 500, "PDF"];
Show[ci, grCDF]
Show[di, grPDF]
```

Out[ ]=



Out[ ]=

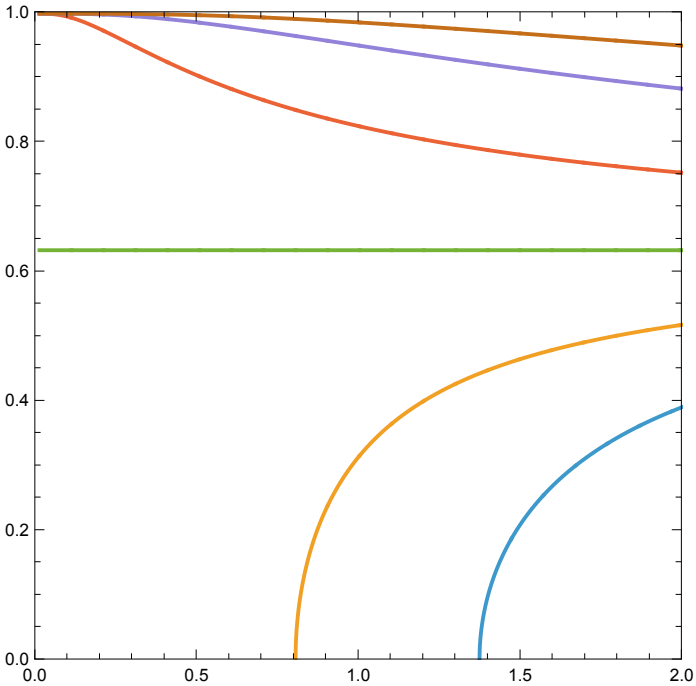


dependencies of sn on the probability that postHC5 > preHC5

In[ ]:=
```
SS = 5;
params = {Kp → K5, n → SS};
del = 0.001;
minSn = sn /. Solve[maxHC5 == z, sn][[1]];
vs = Log10[{2, 1.5, 1, 0.5, 0.1, 0.01}];
mins = {};
Do[
  snnow = minSn /. params /. z → vs[[k]];
  If[snnow < 0.01, AppendTo[mins, 0.01], AppendTo[mins, snnow]];
  , {k, Length[vs]}];
mins
grs = {};
Do[
  snnow = mins[[k]];
  dHC5s = {};
  Do[
   If[snnow > 2.0, Break[]];
   intRng = solX /. params /. z → vs[[k]] /. sn → snnow;
   p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
      CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
   AppendTo[dHC5s, {snnow, p}];
   snnow = snnow + del;
   , {t, 10 000}];
  If[Length[dHC5s] > 0, AppendTo[grs, dHC5s]];
  , {k, Length[vs]}];
ListPlot[grs, Joined → True, AspectRatio → 1,
 Frame → True, PlotRange → {{0, 2}, {0, 1}}]
```

Out[ ]=
```
{1.37027, 0.801557, 0.01, 0.01, 0.01, 0.01}
```

Out[ ]=



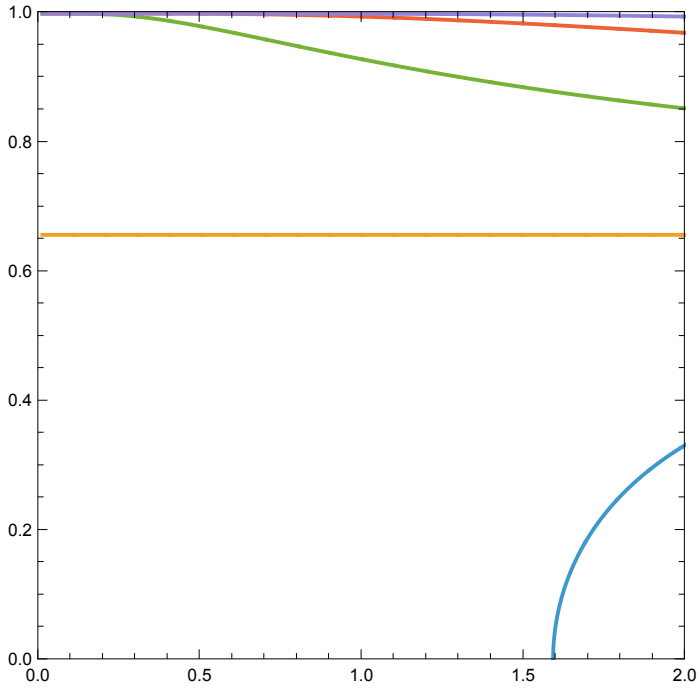
In[ ]:=

```
SS = 10;
params = {Kp → K5, n → SS};
del = 0.001;
minSn = sn /. Solve[maxHC5 == z, sn][[1]];
vs = Log10[{2, 1.5, 1, 0.5, 0.1, 0.01}];
mins = {};
Do[
  snnow = minSn /. params /. z → vs[[k]];
  If[snnow < 0.01, AppendTo[mins, 0.01], AppendTo[mins, snnow]];
  , {k, Length[vs]}];
mins
grs = {};
Do[
  snnow = mins[[k]];
  dHC5s = {};
  Do[
   If[snnow > 2.0, Break[]];
   intRng = solX /. params /. z → vs[[k]] /. sn → snnow;
   p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
      CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
   AppendTo[dHC5s, {snnow, p}];
   snnow = snnow + del;
   , {t, 10 000}];
  If[Length[dHC5s] > 0, AppendTo[grs, dHC5s]];
  , {k, Length[vs]}];
ListPlot[grs, Joined → True, AspectRatio → 1,
 Frame → True, PlotRange → {{0, 2}, {0, 1}}]
```

Out[ ]=

```
{2.71568, 1.58857, 0.01, 0.01, 0.01, 0.01}
```

Out[ ]=



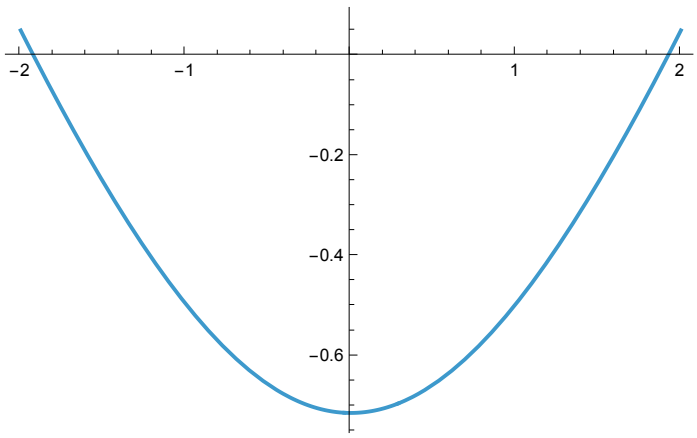
Change in AF

In[ ]:=
```
SS = 5;
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
  knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]}
Plot[DAF[x, SS, Kp, sn, kn, knn] /. params, {x, -2, 2}]
solX = x /. Solve[DAF[x, n, Kp, sn, kn, knn] == z, x]
minAF = DAF[0, n, Kp, sn, kn, knn];

Print["Probability that AF increases"]
intRng = solX /. params /. z → 0;
p = 1 - (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]])
```

Out[ ]=

{Kp → -1.64485, n → 5, sn → 1, kn → 4.20268, knn → 3.70768}

Out[ ]=



Out[ ]=

$$\left\{-\frac{1}{\sqrt{knn^2+2\,knn\,Kp+Kp^2}\;sn}\left(\sqrt{\left(knn^2\,sn^2+2\,knn\,Kp\,sn^2+Kp^2\,sn^2+kn^2\,n\,sn^2-knn^2\,n\,sn^2+2\,kn\,Kp\,n\,sn^2-2\,knn\,Kp\,n\,sn^2+2\,kn\,n\,sn\,z+2\,Kp\,n\,sn\,z+n\,z^2\right)}\right),\right.$$
$$\left.\frac{1}{\sqrt{knn^2+2\,knn\,Kp+Kp^2}\;sn}\left(\sqrt{\left(knn^2\,sn^2+2\,knn\,Kp\,sn^2+Kp^2\,sn^2+kn^2\,n\,sn^2-knn^2\,n\,sn^2+2\,kn\,Kp\,n\,sn^2-2\,knn\,Kp\,n\,sn^2+2\,kn\,n\,sn\,z+2\,Kp\,n\,sn\,z+n\,z^2\right)}\right)\right\}$$

Probability that AF increases

Out[ ]=

0.12723

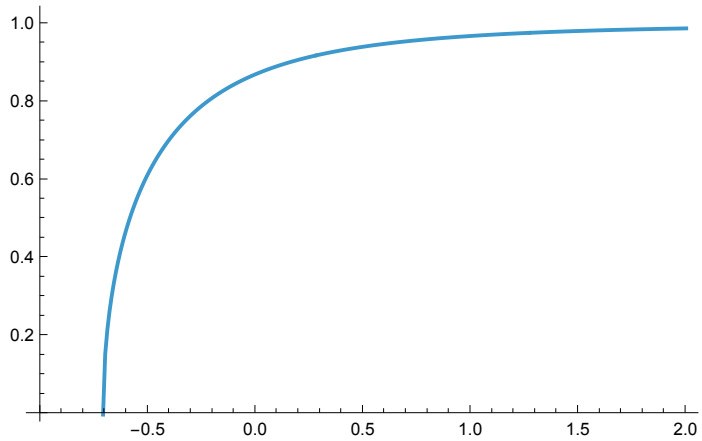
CDF

In[ ]:=
```
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
del = 0.01;
AFnow = minAF /. params
myCDF = {{AFnow, 0}};
Do[
  AFnow = AFnow + del;
  If[AFnow > 2, Break[]];
  intRng = solX /. params /. z → AFnow;
  p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
     CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
  AppendTo[myCDF, {AFnow, p}];
  , {k, 100 000}];
grCDF = ListPlot[myCDF, Joined → True, PlotRange → All, AxesOrigin → {-1, 0}]
```

Out[ ]=

-0.712775

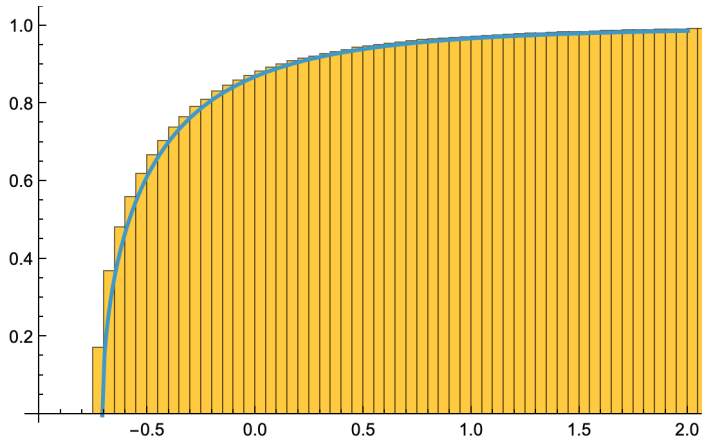
Out[ ]=



comparison with monte-carlo

In[ ]:=
```
dAFs = {};
Do[
  preSample = Table[RandomReal[NormalDistribution[0, 3.14]], {k, SS}];
  preSd = StandardDeviation[preSample];
  preAF = preSd (Kp + kn) /. params;

  postSample = AppendTo[preSample, RandomReal[NormalDistribution[0, 3.14]]];
  postSd = StandardDeviation[postSample];
  postAF = postSd (Kp + knn) /. params;

  dAF = (postAF - preAF) / preSd;
  AppendTo[dAFs, dAF];
  , {t, 10 000}];
ci = Histogram[dAFs, 200, "CDF"];
Show[ci, grCDF, PlotRange → {{-1, 2}, All}, AxesOrigin → {-1, 0}]
```

Out[ ]=



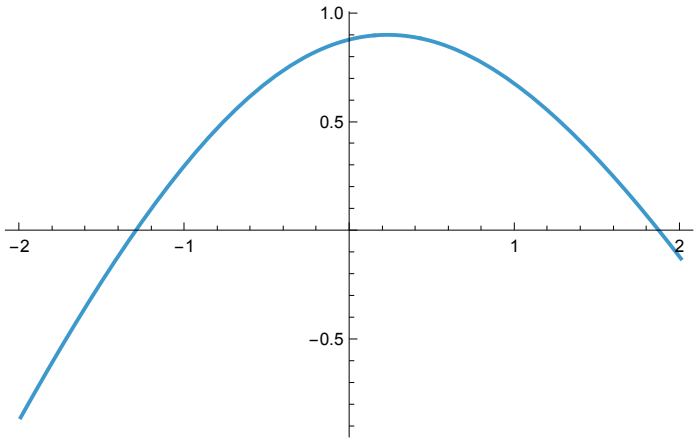
change in PNEC

In[ ]:=
```
SS = 5;
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
  knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]}
Plot[DPNEC[x, SS, Kp, sn, kn, knn] /. params, {x, -2, 2}]
solX = x /. Solve[DPNEC[x, n, Kp, sn, kn, knn] == z, x]
maxX = x /. Simplify[Solve[D[DPNEC[x, n, Kp, sn, kn, knn], x] == 0, x][[2]]]
maxPNEC = DPNEC[maxX, n, Kp, sn, kn, knn]

Print["Probability that PNEC increases"]
intRng = solX /. params /. z → 0;
p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]])
```

Out[ ]=

{Kp → -1.64485, n → 5, sn → 1, kn → 4.20268, knn → 3.70768}

Out[ ]=



Out[ ]=

$$\left\{\frac{1}{2\left(-1+\text{knn}^2+\text{knn}^2\,n\right)\text{sn}^2}\left(-\sqrt{n\,(1+n)}\;\text{sn}\,(-2\,\text{kn}\,\text{sn}+2\,z)-\sqrt{\left(n\,(1+n)\,\text{sn}^2\,(-2\,\text{kn}\,\text{sn}+2\,z)^2-4\left(-1+\text{knn}^2+\text{knn}^2\,n\right)\text{sn}^2\left(-\text{knn}^2\,\text{sn}^2-\text{kn}^2\,n\,\text{sn}^2-\text{kn}^2\,n^2\,\text{sn}^2+\text{knn}^2\,n^2\,\text{sn}^2+2\,\text{kn}\,n\,\text{sn}\,z+2\,\text{kn}\,n^2\,\text{sn}\,z-n\,z^2-n^2\,z^2\right)\right)}\right),\right.$$
$$\left.\frac{1}{2\left(-1+\text{knn}^2+\text{knn}^2\,n\right)\text{sn}^2}\left(-\sqrt{n\,(1+n)}\;\text{sn}\,(-2\,\text{kn}\,\text{sn}+2\,z)+\sqrt{\left(n\,(1+n)\,\text{sn}^2\,(-2\,\text{kn}\,\text{sn}+2\,z)^2-4\left(-1+\text{knn}^2+\text{knn}^2\,n\right)\text{sn}^2\left(-\text{knn}^2\,\text{sn}^2-\text{kn}^2\,n\,\text{sn}^2-\text{kn}^2\,n^2\,\text{sn}^2+\text{knn}^2\,n^2\,\text{sn}^2+2\,\text{kn}\,n\,\text{sn}\,z+2\,\text{kn}\,n^2\,\text{sn}\,z-n\,z^2-n^2\,z^2\right)\right)}\right)\right\}$$

Out[ ]=

$$\frac{\sqrt{-1+n}}{\sqrt{-1+\text{knn}^2\,(1+n)}}$$

Out[ ]=

$$-\left(\left(-kn-Kp+(knn+Kp)\sqrt{\frac{-1+n}{n}+\frac{-1+n}{n\left(-1+knn^2\,(1+n)\right)}}\right)sn\right)+\left(\frac{\sqrt{-1+n}}{\sqrt{n\,(1+n)}\,\sqrt{-1+knn^2\,(1+n)}}+Kp\left(-1+\sqrt{\frac{-1+n}{n}+\frac{-1+n}{n\left(-1+knn^2\,(1+n)\right)}}\right)\right)sn$$

Probability that PNEC increases

Out[ ]=

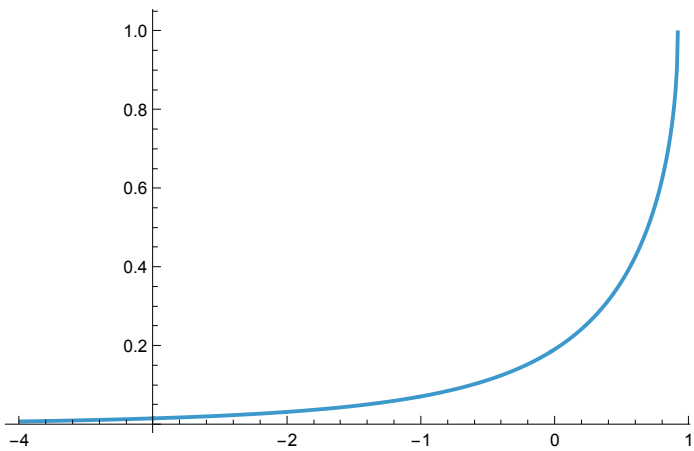
0.802623

CDF

In[ ]:=

```
SS = 5;
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
maxZ = maxPNEC /. params;
del = 0.01;
znow = -4;
myCDF = {};
Do[
 intRng = solX /. params /. z → znow;
 If[znow > maxZ, Break[]];
 p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
 AppendTo[myCDF, {znow, 1 - p}];
 znow = znow + del;
 , {k, 100 000}]
myCDF = AppendTo[myCDF, {maxZ, 1}];
grCDF = ListPlot[myCDF, Joined → True, PlotRange → All, AxesOrigin → {-3, 0}]
```

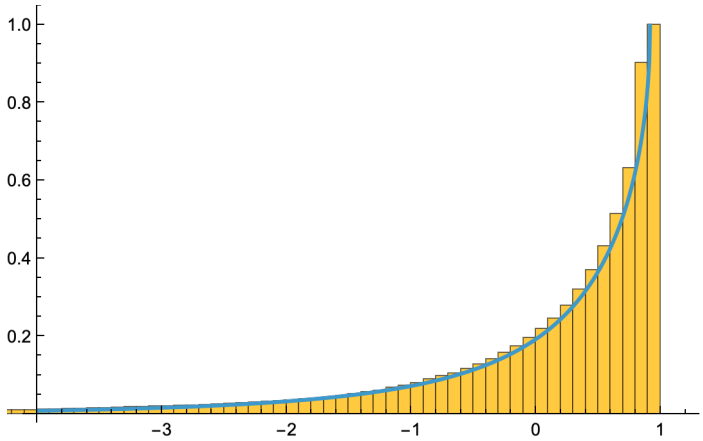
Out[ ]=



comparison with monte-carlo

In[ ]:=
```
dPNECs = {};
Do[
  preSample = Table[RandomReal[NormalDistribution[0, 3.14]], {k, SS}];
  preMean = Mean[preSample];
  preSd = StandardDeviation[preSample];
  preNOEC = preMean - (kn preSd /. params);

  postSample = AppendTo[preSample, RandomReal[NormalDistribution[0, 3.14]]];
  postMean = Mean[postSample];
  postSd = StandardDeviation[postSample];
  postNOEC = postMean - (knn postSd /. params);

  dPNEC = (postNOEC - preNOEC) / preSd;
  AppendTo[dPNECs, dPNEC];
  , {t, 10 000}];
ci = Histogram[dPNECs, 200, "CDF"];
Show[ci, grCDF, PlotRange → {{-4, 1.2}, All}, AxesOrigin → {-4, 0}]
```

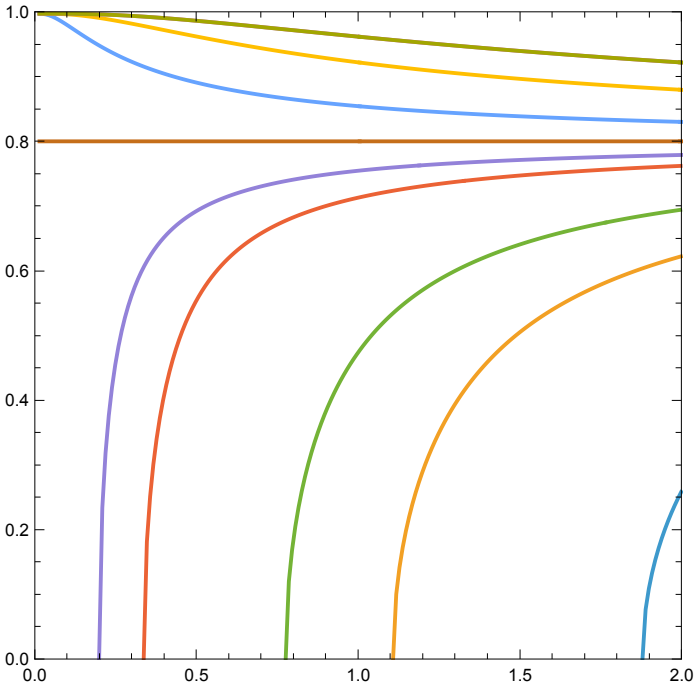
Out[ ]=



dependencies of sn on the probability that postPNEC > v prePNEC

In[ ]:=
```
SS = 5;
params = {Kp → K5, n → SS, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
del = 0.01;
minSn = sn /. Solve[maxPNEC == z, sn][[1]];
snnow = minSn /. params /. z → vs[[2]];

vs = Log10[{50, 10, 5, 2, 1.5, 1, 0.5, 0.1, 0.01, 0.01}];

mins = {};
Do[
  snnow = minSn /. params /. z → vs[[k]];
  If[snnow < 0.01, AppendTo[mins, 0.01], AppendTo[mins, snnow]];
  , {k, Length[vs]}];
grs = {};
Do[
  snnow = mins[[k]];
  dPNECs = {};
  Do[
   If[snnow > 2.0, Break[]];
   intRng = solX /. params /. z → vs[[k]] /. sn → snnow;
   p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
      CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
   AppendTo[dPNECs, {snnow, p}];
   snnow = snnow + del;
   , {t, 10 000}];
  If[Length[dPNECs] > 0, AppendTo[grs, dPNECs]];
  , {k, Length[vs]}];
ListPlot[grs, Joined → True, AspectRatio → 1,
 Frame → True, PlotRange → {{0, 2}, {0, 1}}]
```

Out[ ]=

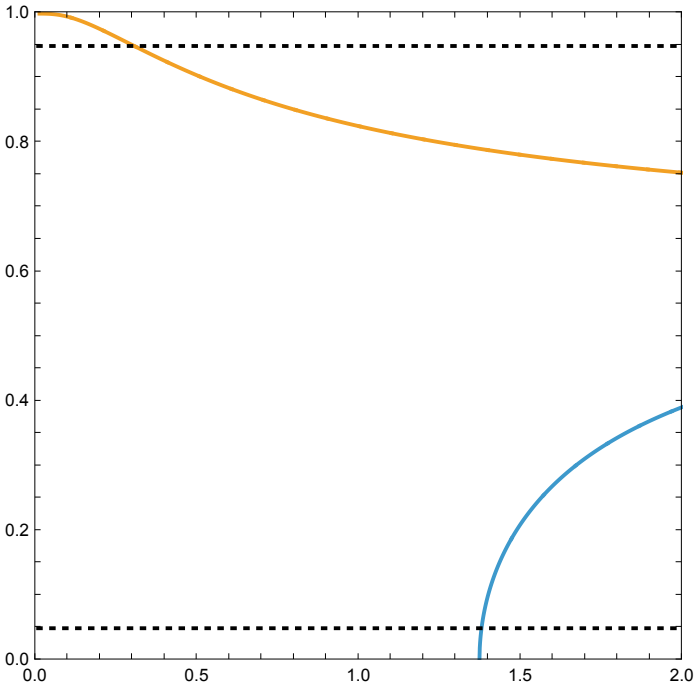


Values of sn for which P(post-HC5 <½× or >2× pre-HC5) = 0.05

This involves finding the intersection of the wavy line and the curve in the diagram below.
Blue lines is for postHC5 > 2 preHC5, and orange line is for postHC5 > 1/2 preHC5.
As is clear from the figure, sn for postHC5 > 1/2 preHC5 is lower, so we consider only this case.

```
In[ ]:= SS = 5;
solX = FullSimplify[x /. Solve[DHC5[x, n, Kp, sn] == z, x]];
maxX = FullSimplify[x /. Solve[D[DHC5[x, n, Kp, sn], x] == 0, x][[2]]];
maxHC5 = FullSimplify[DHC5[maxX, n, Kp, sn]];
params = {Kp → K5, n → SS};
del = 0.001;
minSn = sn /. Solve[maxHC5 == z, sn][[1]];
vs = Log10[{2, 0.5}];
mins = {};
Do[
  snnow = minSn /. params /. z → vs[[k]];
  If[snnow < 0.01, AppendTo[mins, 0.01], AppendTo[mins, snnow]];
  , {k, Length[vs]}];
grs = {};
Do[
  snnow = mins[[k]];
  dHC5s = {};
  Do[
   If[snnow > 2.0, Break[]];
   intRng = solX /. params /. z → vs[[k]] /. sn → snnow;
   p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
      CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
   AppendTo[dHC5s, {snnow, p}];
   snnow = snnow + del;
   , {t, 10 000}];
  If[Length[dHC5s] > 0, AppendTo[grs, dHC5s]];
  , {k, Length[vs]}];
ai = ListPlot[grs, Joined → True,
   AspectRatio → 1, Frame → True, PlotRange → {{0, 2}, {0, 1}}];
bi = Plot[0.95, {x, 0, 2.0}, PlotStyle → {Dashing[{0.01, 0.01}], GrayLevel[0]}];
ci = Plot[0.05, {x, 0, 2.0}, PlotStyle → {Dashing[{0.01, 0.01}], GrayLevel[0]}];
Show[ai, bi, ci]
```

Out[ ]=

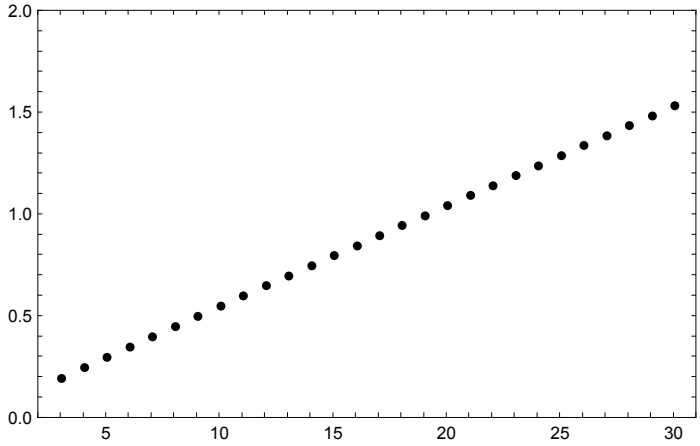


Define a function to calculate the probability for any sn

In[ ]:=
```
pCalc05[solX_, params_, maxHC5_, snv_, val_] := Module[{p},
   intRng = solX /. params /. z → Log10[val] /. sn → snv;
   p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
      CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
   Return[p];
  ];
```

In[ ]:=
```
(*Pr[postHC5>1/2 preHC5] = 0.05: Pr[postHC5 < 1/2 preHC5] = 0.95  *)
(*newton method*)
del = 0.0001;
sns = {};
Do[
  params = {Kp → K5, n → k};
  snnow = 1.;
  Do[
   FNow = pCalc05[solX, params, maxHC5, snnow, 0.5] - 0.95;
   FAft = pCalc05[solX, params, maxHC5, snnow + del, 0.5] - 0.95;
   If[Abs[FNow] < 0.0000001, Break[]];
   grad = (FAft - FNow) / del;
   snnow = snnow - 0.01 FNow / grad;
   , {t, 10 000}];
  AppendTo[sns, {k, snnow}];
  , {k, 3, 30}];
ListPlot[sns, PlotStyle → {GrayLevel[0]},
 Frame → True, PlotRange → {{2, 31}, {0, 2}}]
```

Out[ ]=



Check it is really 0.95

In[ ]:=
```
Do[
 params = {Kp → K5, n → sns⟦k⟧⟦1⟧, sn → sns⟦k⟧⟦2⟧};
 intRng = solX /. params /. z → Log10[0.5];
 p = (CDF[StudentTDistribution[SS - 1], intRng⟦2⟧] -
    CDF[StudentTDistribution[SS - 1], intRng⟦1⟧]);
 Print["sample size = ", sns⟦k⟧⟦1⟧, ", probability = ", p];
 , {k, Length[sns]}]
```

```
sample size = 3, probability = 0.95
sample size = 4, probability = 0.95
sample size = 5, probability = 0.95
sample size = 6, probability = 0.95
sample size = 7, probability = 0.95
sample size = 8, probability = 0.95
sample size = 9, probability = 0.95
sample size = 10, probability = 0.95
sample size = 11, probability = 0.95
sample size = 12, probability = 0.95
sample size = 13, probability = 0.95
sample size = 14, probability = 0.95
sample size = 15, probability = 0.95
sample size = 16, probability = 0.95
sample size = 17, probability = 0.95
sample size = 18, probability = 0.95
sample size = 19, probability = 0.95
sample size = 20, probability = 0.95
sample size = 21, probability = 0.95
sample size = 22, probability = 0.95
sample size = 23, probability = 0.95
sample size = 24, probability = 0.95
sample size = 25, probability = 0.95
sample size = 26, probability = 0.95
sample size = 27, probability = 0.95
sample size = 28, probability = 0.95
sample size = 29, probability = 0.95
sample size = 30, probability = 0.95
```

Cadmium toxicity by Van Straalen and Dennneman (1989)

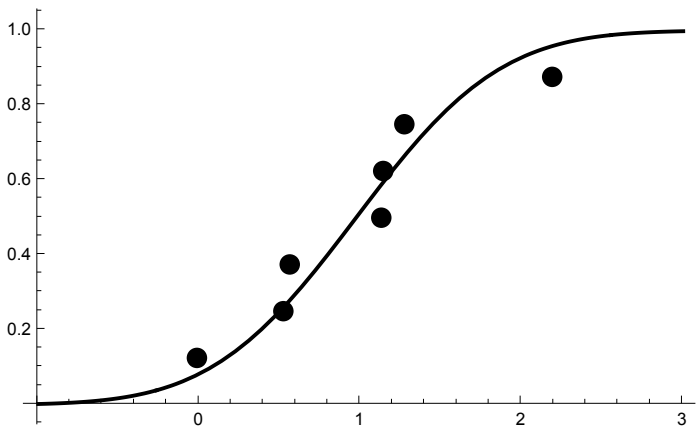
In[ ]:=
```
NOECs = Log10[{0.97, 3.33, 3.63, 13.50, 13.80, 18.70, 154.}]
ya = Table[k / 8, {k, 7}];
m = Mean[NOECs]
sd = StandardDeviation[NOECs]
ai = Plot[CDF[NormalDistribution[m, sd], x],
   {x, -1, 3}, PlotStyle → GrayLevel[0], AxesOrigin → {-1, 0}];
bi = ListPlot[Transpose[{NOECs, ya}],
   PlotStyle → {GrayLevel[0], PointSize[0.03]}];
Show[ai, bi]
```

Out[ ]=
```
{-0.0132283, 0.522444, 0.559907, 1.13033, 1.13988, 1.27184, 2.18752}
```

Out[ ]=
```
0.971243
```

Out[ ]=
```
0.702761
```

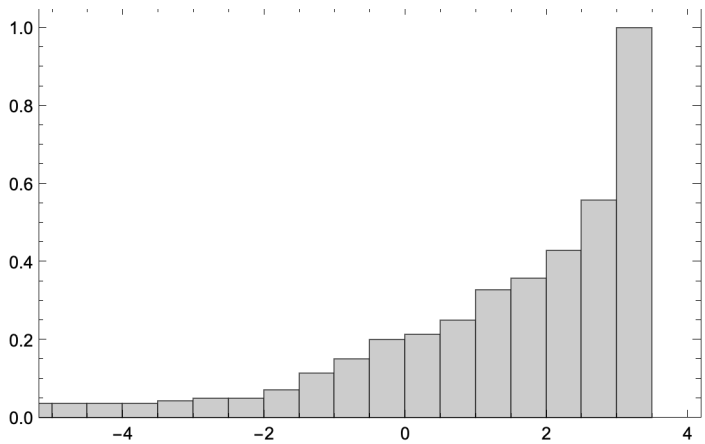
Out[ ]=




```
(* Take 3 NOECs from the original datase, and add one*)
```

In[ ]:=
```
SS = 3;
params = {Kp → K5, n → SS, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
dNOECs = {};
Do[
  sLst = IntegerDigits[k, 2, 7];
  num = uvec.sLst;
  If[num == SS, {
    sNOECs = {};
    nNOECs = {};
    Do[
     If[sLst[[t]] == 1, AppendTo[sNOECs, NOECs[[t]]], AppendTo[nNOECs, NOECs[[t]]]];
     , {t, 7}];

    preMean = Mean[sNOECs];
    preSD = StandardDeviation[sNOECs];
    preNOEC = preMean - (kn /. params) preSD;

    Do[
     aNOECS = Flatten[{sNOECs, nNOECs[[j]]}];
     postMean = Mean[aNOECS];
     postSD = StandardDeviation[aNOECS];
     postNOEC = postMean - (knn /. params) postSD;
     AppendTo[dNOECs, (postNOEC - preNOEC) / preSD];
     , {j, 7 - SS}];
   }];
  , {k, 1, tnum}];
grHist = Histogram[dNOECs, 75, "CDF", PlotRange → {{-5, 4}, {0, 1}},
  AxesOrigin → {-5, 0}, Frame → True, Frame → True, ChartStyle → GrayLevel[0.8]]
```

Out[ ]=

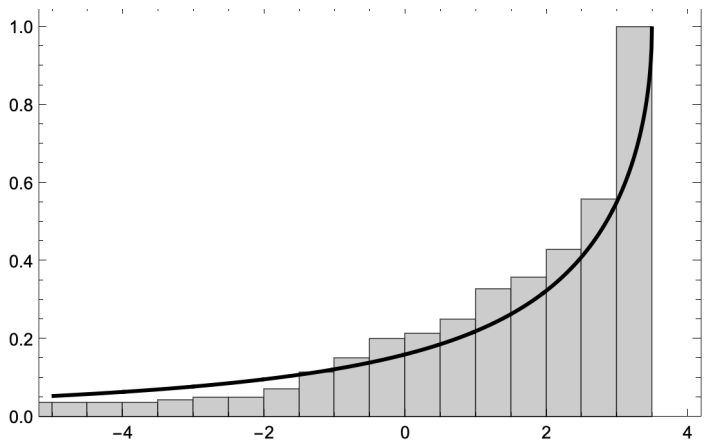


In[ ]:=
```
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
solX = x /. Solve[DPNEC[x, n, Kp, sn, kn, knn] == z, x];
maxX = x /. Simplify[Solve[D[DPNEC[x, n, Kp, sn, kn, knn], x] == 0, x][[2]]];
maxPNEC = DPNEC[maxX, n, Kp, sn, kn, knn];
maxZ = maxPNEC /. params
del = 0.01;
znow = -5;
myCDF = {};
Do[
 intRng = solX /. params /. z → znow;
 If[znow > maxZ, Break[]];
 p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
 AppendTo[myCDF, {znow, 1 - p}];
 znow = znow + del;
 , {k, 100 000}]
myCDF = AppendTo[myCDF, {maxZ, 1}];
grCDF = ListPlot[myCDF, Joined → True,
   PlotRange → All, AxesOrigin → {-3, 0}, PlotStyle → GrayLevel[0]];
Show[grHist, grCDF]
```

Out[ ]=

3.47558

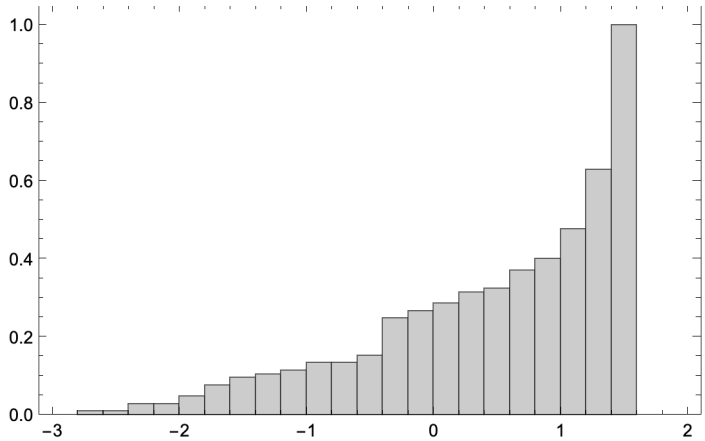
Out[ ]=




```
(* Take 3 NOECs from the original datase, and add one*)
```

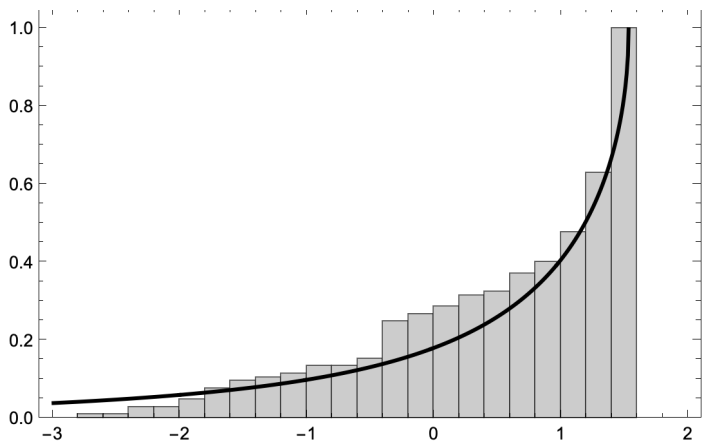
In[ ]:=
```
SS = 4;
params = {Kp → K5, n → SS, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
dNOECs = {};
Do[
  sLst = IntegerDigits[k, 2, 7];
  num = uvec.sLst;
  If[num == SS, {
    sNOECs = {};
    nNOECs = {};
    Do[
     If[sLst[[t]] == 1, AppendTo[sNOECs, NOECs[[t]]], AppendTo[nNOECs, NOECs[[t]]]];
     , {t, 7}];

    preMean = Mean[sNOECs];
    preSD = StandardDeviation[sNOECs];
    preNOEC = preMean - (kn /. params) preSD;

    Do[
     aNOECS = Flatten[{sNOECs, nNOECs[[j]]}];
     postMean = Mean[aNOECS];
     postSD = StandardDeviation[aNOECS];
     postNOEC = postMean - (knn /. params) postSD;
     AppendTo[dNOECs, (postNOEC - preNOEC) / preSD];
     , {j, 7 - SS}];
   }];
  , {k, 1, tnum}];
grHist = Histogram[dNOECs, 20, "CDF", PlotRange → {{-3, 2}, {0, 1}},
  AxesOrigin → {-3, 0}, Frame → True, ChartStyle → GrayLevel[0.8]]
```

Out[ ]=



In[ ]:=
```
params = {Kp → K5, n → SS, sn → 1, kn → Quantile[
      NoncentralStudentTDistribution[SS - 1, -Sqrt[SS] K5], 0.95] / Sqrt[SS],
   knn → Quantile[
      NoncentralStudentTDistribution[SS, -Sqrt[SS + 1] K5], 0.95] / Sqrt[SS + 1]};
maxZ = maxPNEC /. params
del = 0.01;
znow = -3;
myCDF = {};
Do[
 intRng = solX /. params /. z → znow;
 If[znow > maxZ, Break[]];
 p = (CDF[StudentTDistribution[SS - 1], intRng[[2]]] -
    CDF[StudentTDistribution[SS - 1], intRng[[1]]]);
 AppendTo[myCDF, {znow, 1 - p}];
 znow = znow + del;
 , {k, 100 000}]
myCDF = AppendTo[myCDF, {maxZ, 1}];
grCDF = ListPlot[myCDF, Joined → True,
   PlotRange → {{-3, 2}, {0, 1}}, PlotStyle → GrayLevel[0], Frame → True];
Show[grHist, grCDF, PlotRange → {{-3, 2}, {0, 1}}]
```

Out[ ]=

1.52522

Out[ ]=